

IX. MINI PROJECT – SMART AGRICULTURE DATA ANALYSIS

AIM:

To clean, process, analyze, and visualize smart agriculture data using Python, and to build interactive dashboards using Power BI and Tableau to derive meaningful insights about crop yield, rainfall, temperature, soil conditions, crop types, and agricultural productivity.

DOMAIN:

Agriculture Data Analytics

PROBLEM STATEMENT:

Agricultural systems generate large amounts of data containing information about crop yield, rainfall, temperature, soil conditions, crop types, and agricultural production. It is difficult to identify high-yield crops, region-wise agricultural performance, weather patterns, and factors affecting crop production from raw data.

Therefore, there is a need for a data-driven analytics system that cleans, analyzes, and visualizes agriculture data to extract meaningful insights about crop productivity and environmental factors.

OUTPUT:

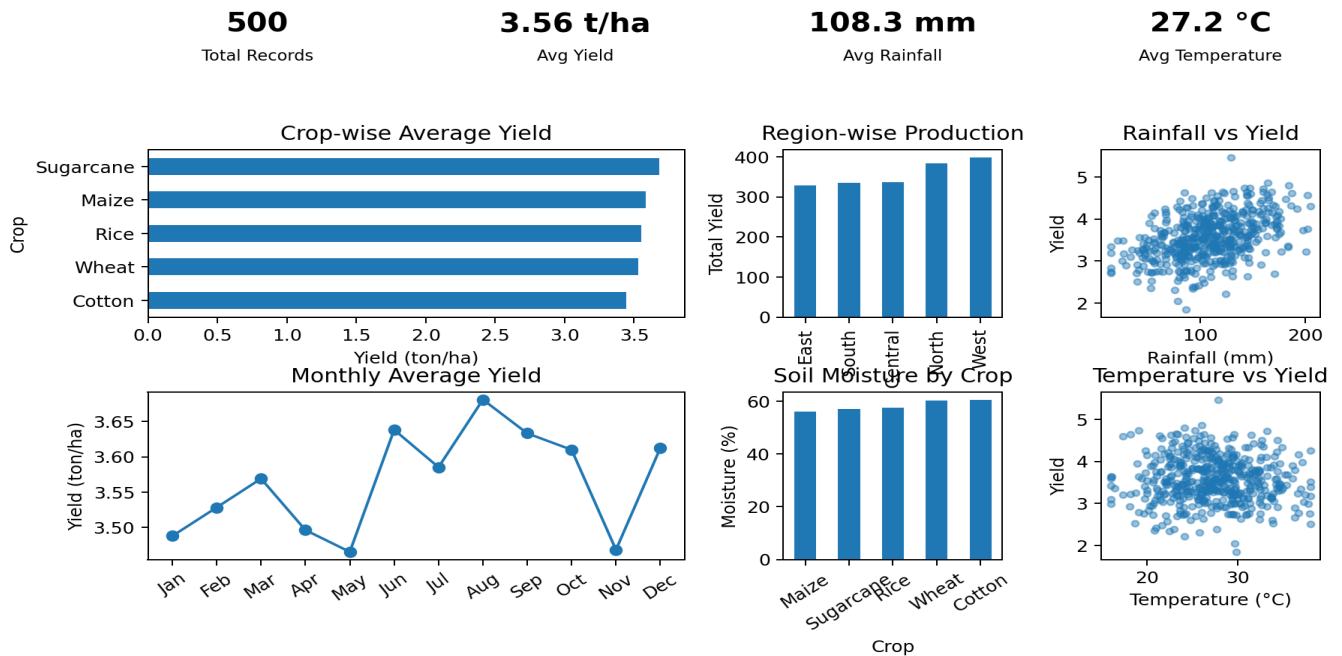
Python:

- Crop-wise Average Yield
- Region-wise Crop Production
- Rainfall vs Crop Yield
- Temperature Trends
- Soil Condition vs Crop Yield
- Crop Price Trends

Power BI:

The smart agriculture data is represented using an interactive Power BI dashboard.

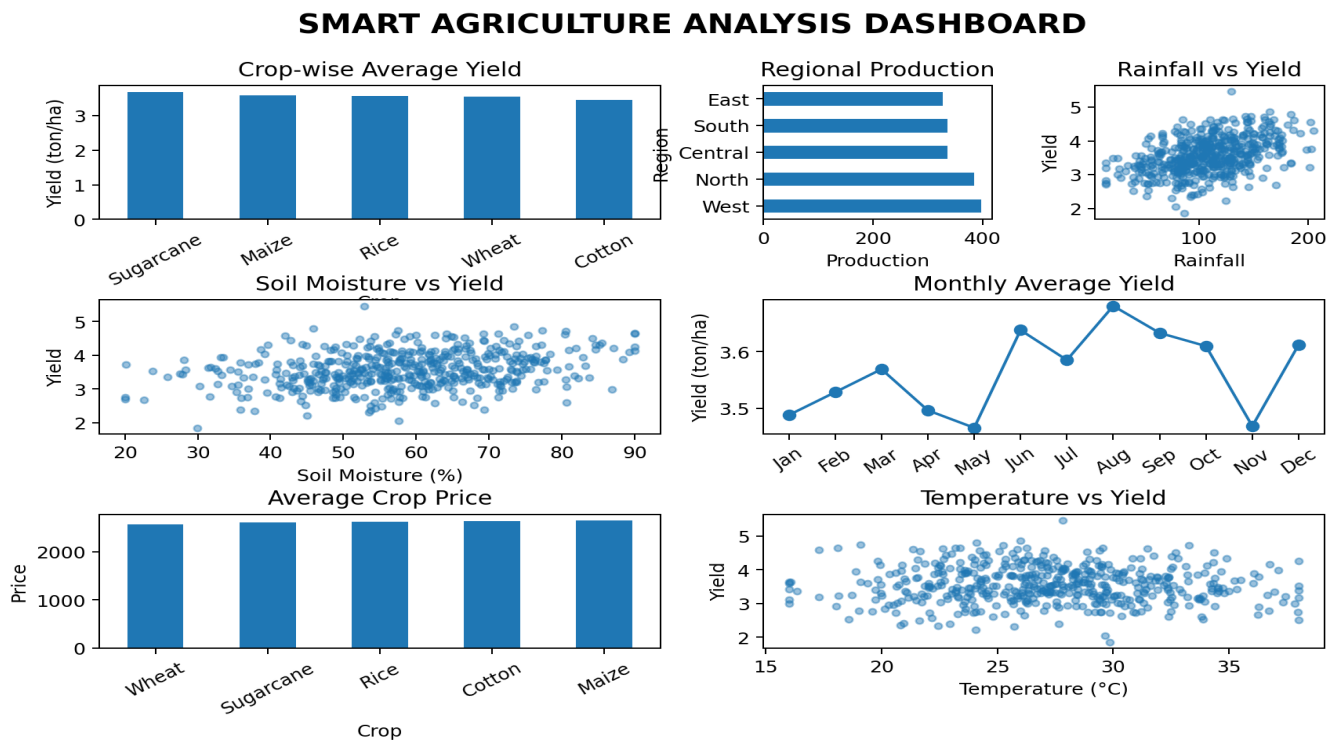
SMART AGRICULTURE DATA ANALYSIS DASHBOARD



The dashboard contains total agricultural records, average crop yield, crop-wise performance, region-wise production, rainfall versus yield, monthly yield performance, soil moisture analysis, temperature trends, and interactive filters.

Tableau:

The same smart agriculture dataset is visualized using an interactive Tableau dashboard.



The Tableau dashboard presents crop-wise yield, region-wise production, rainfall versus crop yield, soil moisture versus yield, monthly performance, crop price trends, temperature versus yield, and filter options.

RESULT:

The Smart Agriculture Data Analysis project was successfully implemented using Python, Power BI, and Tableau. The dataset was analyzed and transformed into static and interactive visualizations, providing meaningful insights into crop yield, agricultural production, rainfall, temperature, soil conditions, crop types, and crop prices.